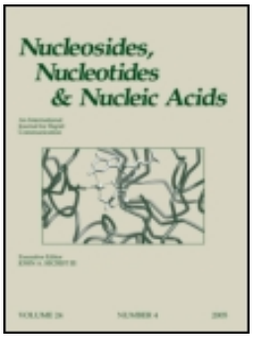

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DISCOVERY OF CENTROSOMAL RNA AND CENTROSOMAL HYPOTHESIS OF CELLULAR AGEING AND DIFFERENTIATION

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□ *In 2006, a group of scientists studying centrosomes of* Spisula solidissima *mollusc oocytes under the leadership of Alliegro (Alliegro, M.C.; Alliegro, M.A.; Palazzo, R.E. Centrosome-associated RNA in surf clam oocytes.* Proc. Natl. Acad. Sci. USA ***2006****, 103(24), 9034–9038) reliably demonstrated the existence of specific RNA in centrosome, called centrosomal RNA (cnRNA). In their first article, five different RNAs (cnRNAs 11, 102, 113, 170, and 184) were described. During the process of full sequencing of the first transcript (cnRNA 11), it was discovered that the transcript contained a conserved structure—a reverse transcriptase domain located together with the most important centrosomal protein, γ-tubulin. In an article published in 2005, we made assumptions about several possible mechanisms for determining the most important functions of centrosomal structures and referred to one of them as a "RNA-dependent mechanism." This idea about participation of hypothetic centrosomal small interference RNA and/or microRNA in the process was made one year prior to the discovery of cnRNA by Alliegro's group. The discovery of specific RNA in a centrosome is indirect evidence of a centrosomal hypothesis of cellular ageing and differentiation. The presence of a reverse transcriptase domain in this type of RNA, together with its uniqueness and specificity, makes the centrosome a place of information storage and reproduction.*

Keywords Centrosome; centrioles; ageing; centrosomal RNA

INTRODUCTION

Six years have passed since the first publication of our centrosomal (centriolar) hypothesis of cell differentiation and ageing.[1] In this article, we will be discussing new studies relating to centrosomal RNA (cnRNA), including research on the discovery of cnRNA, in the light of this hypothesis. There is

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a need for strengthening the experimental base of the centrosomal hypothesis of cellular ageing and differentiation;[2,3] however, this problem will be discussed in more detail in this article.

The possible presence of RNA in the centrosome as one of the mechanisms of centrosomal influence on a cell and a mechanism of information storage and reproduction has already been discussed.[1,4] The assumption was based on the following consideration: Because there are only four self-duplicating cellular organelles, namely nucleus, mitochondria, plastids, and centrosome,[5] and the first three of these have their own DNA, it would also be logical to assume the presence of DNA/RNA in the centrosome. However, even in 1970 the consensus was that centrosomes do not contain DNA.[6] Consequently, one might assume the presence in centrosomes of RNA only.

mRNA and the Search for Nucleic Acids in the Centrosome

The first data on the presence of nucleic acids in a centrosome were put forward in the mid-20th century. Although it was impossible to provide reliable evidence, this idea did not lose its attractiveness in the light of the theory of Margulis about the origin of centrosome from a spirochete ancestor (cited by Chapman et al.[7]). Therefore, scientists continued to look for nucleic acids in the 1970s and 1980s. Many studies claimed to have found RNA in centrosomes or centrosome-related material,[8–10] but proof of its existence was lacking because of the low sensitivity of the methods applied during that period.

The role of mRNA localization in various cellular functions was discovered in 1983[11]; only after the accumulation of a significant volume of experimental material did it become clear that this plays an important role in most cellular functions.[12] It also transpired that different cytoskeletal structures play a leading role in mRNA anchoring.[11,13–16] Microtubules play a leading role in the formation of cytoskeleton,[17,18] and in turn they are regulated by the centrosome acting as a major microtubule organization center or MTOC.[19,20] That is why interest has arisen in the microtubular structures and centrosome regarding the location of mRNA anchoring. Many interesting works have since been undertaken in this direction, and in particular the study of *Ilyanassa obsolete* embryos showed that asymmetric distribution of many maternal cytoplasmic factors takes place during division. These factors include transcripts of developmental genes, e.g. *even-skipped (Eve), decapentaplegic (Dpp)*, and *tolloid (Tld)*, that participate in embryonic pattern formation. During interphase, these are transported to centrosomes by microtubules, ensuring asymmetry of their distribution in sister blastomeres, which eventually brings about differential activity of genes.[21] Kingsley et al.[22] continued this study and found that centrosomes were used as specific "way-stations," which is a common phenomenon in *Ilyanassa* embryos. They also discovered

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that many centrosome-related transcripts are unique and absent from the known databases. Based on their own results and the experiments of others, these authors concluded that the process of *microRNA* (mRNA) distribution in the cells means that the fate of these was in some part defined by the inheritance of centrosomally localized mRNAs.

In embryonic cells of Drosophila, Lecuyer et al.[12] identified six RNAs of centrosomal localization and made the assumption that they participated in the centrosome assemblage process. Blower et al.[23] (2007) noted in particular a quite specific and conservative group of mRNAs related to microtubular structures in *Xenopus laevis*, which they called microtubule-bound mRNA (MT-mRNA). Previously, Blower et al.[24] had explained the necessity of such a connection between specific RNAs and the spindle on the following considerations:

1. Interaction with the spindle allows local translation of specific mitotic regulators.
2. Asymmetry of cellular division is ensured by the binding of specific mRNAs to centrosome.
3. RNAs play a structural role in the spindle apparatus assemblage process.

Therefore, the conclusion was reached that local restriction of protein synthesis by means of *mRNA* localization is the most important mechanism in the establishment of asymmetric distribution of cytoplasm factors, which is encountered in many cell types.[11,15,25]

However, none of the above-mentioned authors have claimed to have discovered a unique type of RNA related to microtubules/spindles or division/centrosome; as a rule, they were considered part of general cellular mRNAs. Kingsley et al.[22] provided the exception by indicating that some of the mRNAs they discovered were unique and exclusively (or almost exclusively) connected with centrosomes.

CENTROSOMAL RNA

A group studying the centrosomes of *Spisula solidissima* mollusc oocytes demonstrated the existence of specific RNA, referred to as cnRNA.[26] This discovery was after making use of large centrosomes in oocytes and zygotes of this type of mollusc, finding a reliable and clean method for their isolation,[27] and with a clear knowledge of precise mechanisms of development of this structure—in particular, its absence in unfertilized oocytes and assemblage at the 4th min after activation.[28] They isolated RNA directly from the purified centrosome preparations, amplifying it by RT-PCR and cloning it to create a library of cnRNAs. In their first article, five different RNAs

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(cnRNAs 11, 102, 113, 170, and 184) were described. Their size varied between 638 and 869 base pairs, and they were unique transcripts from the family characteristic only of the centrosome. These RNAs are present only in meager amounts in the cell. Unlike the above-mentioned transcripts of developmental genes, *Eve*, *DPP*, and *TlD* that *Ilyanassa obsoleta* snail blastomeres contain, they could not be detected by Northern analysis, even by using RT-PCR, when analysis was carried out on whole cell RNA. Thus, the amount of this RNA in the general cytoplasm must be almost vanishingly small. Comprehensive analysis of five clones showed that four of these did not have reliable correspondence either in DNA or protein databases.[26]

Alliegro and co-workers continued to isolate and identify RNA from the centrosome, and to create a full library of cnRNAs.[29] They described 120 different clones created from association with cnRNA, and only one-third (39/120) had features similar to sequences described in the database. In this article, the authors also tried to find DNA corresponding to cnRNAs in oocytes, spermatozoids, and isolated germinal vesicles. PCR analysis indicated that cnRNA sequences are present in the DNA obtained from the above-mentioned structures (this somewhat contradicts the data from the first article of this group). But it turned out that they contain practically no introns, and this makes them similar to the DNA of viruses and bacteria rather than *Metazoa*.[5]

During full sequencing of the first transcript (cnRNA 11), it was found to contain a conservative structure—a reverse transcriptase domain—located together with the most important centrosomal protein, γ-tubulin.[26] Two other cnRNAs (cnRNA15 and cnRNA194) were fully sequenced subsequently.[29] BLAST analysis showed that cnRNA15 also contains a conservative reverse transcriptase domain similar to endonuclease/reverse transcriptase of sea urchin, *Strongylocentrotus purpuratus*. According to these researchers, the translation mechanism of cnRNA15 and cnRNA194 translation is unknown. However, they think that there are four explanations for the presence of RNA in the centrosome, namely it is (1) an artifact related to adhesion of cytoplasmic transcripts to the centrosome during cell fractionation; (2) an artifact related to contamination during separation of isolated centrosome; and (3) an artifact resulting from infection of the oocyte with a virus (during aggregation of viral particles delivered by means of microtubular transport). However, these considerations were refuted by detailed in situ hybridization and BLAST analysis. Undertaking such an analysis was important in ensuring that the discovery of a new type of RNA was valid. Hence, only the fourth consideration remained, i.e. cnRNA represents a class or a family of molecules truly related to centrosome.[29]

Irrespective of the fact that within a month from the publication of Alliegro and co-authors'[26] first report, the discovery was highlighted in an article by Pederson,[30] the only scientist working on the problems of symbiogenesis who was interested in this study. A number of researchers believed

that it confirmed the notion of Margulis et al.[31] of the symbiotic origin of centrosomal/cytoskeletal structures. However, the origin of centrioles and centrosomes was seen by Chapman and Alliegro[28] as different from that of the viral hypothesis of Margulis. In a subsequent article, Alliegro and Satir[32] provided a convincing substantiation of the idea of viral origin of centrioles and centrosome. They believed that apart from earlier arguments, the symbiotic origin of centrosome/centrioles is corroborated by the following facts: A small amount of cnRNAs have similarities in protein and nucleic acid databases; the presence of a very limited amount of cytoplasmic mRNA among centrosome-related ribonucleic structures; an extremely low content of introns in cnRNAs-related genes of the nucleus (unlike the same cytoplasmic mRNA); and the presence in the majority of fully sequenced cnRNA structures similar to reverse transcriptase. These authors provide additional arguments in favor of their hypothesis. In particular, they have analyzed the relatedness of cnRNAs with strong centrosomal localization, as well as relatedness of centrosomal/centriolar proteins to molecules of viruses. For the evaluation of hits between these structures, they have used quite severe e-value (e^{-5} to e^{-64}) criteria. It transpired that many unique centrosomal and centriolar proteins had a high level of correspondence with viral proteins. Out of an aggregate eight centrosome-associated RNAs reported by two independent laboratories,[22,26,29] six have similarity to entries in viral databases, all but one with e-values of $\leq 1e^{-5}$. It should be noted that there is no clear guideline for a cutoff e-value to demonstrate relationship by descent (Table 1). Some investigators accept values of $\leq 1e^{-2}$ as indicating relatedness; others prefer the more rigorous $1e^{-5}$.[32] It is also interesting that such a close relationship was revealed between centrosomal/centriolar contents and viruses, but not between cytoskeletal proteins and viruses (as the authors state, "some centrosomal and centriolar molecules clearly have relatives among the viruses that meet or exceed criteria conventionally accepted to indicate a relationship by descent" (ref [32], p. 62)). In contrast,

TABLE 1 Similarity between *centrosome*-associated *RNA*s and viral sequences (from ref. [32], with modifications)

Query RNA	Accession	Viral hit	E-value
Nanos-like protein	EU087572	Cyprinid herpes virus 3 DNAb	$6.0e^{-5}$
Ankyrin-like protein	EU087604	Paramecium chlorella virus hypothetical protein	$7.0e^{-18}$
Lipoprotein VsaC-like protein	EU087576	Crocodilepox virus virion core protein	$3.0e^{-10}$
Zinc finger family 135 protein	EU08758	5 Human APM-1 protein	$9.0e^{-13}$
cnRNA11	DQ359732	Baculo virus retrotransposón	$3.0e^{-64}$
cnRNA15	EU069824	Lymphocystis virus RNA-dependent DNA polymerase	$8.0e^{-7}$

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out of 10 cytoskeletal proteins examined, no significant similarity with viral particles was found in nine.[32] Based on these facts, the researchers assumed that the appearance of centrosome and/or centriole in the cell resulted from a long-standing viral symbiogenesis, whereas the cytoskeleton apparently has a non-viral origin.

POSSIBLE FUNCTIONS OF cnRNA

Reviewing studies on the topic "centrosomal structures and RNA," we wished to determine whether the findings might substantiate our own centrosomal (centriolar) hypothesis of cellular ageing and differentiation. Here we give a brief formulation of the main arguments. Centrosome (centriole) and related structures (cytoskeleton) determine processes of differentiation and age-related changes in multicellular animal cells.[1,4] We decided to relate replicative ageing of dividing cells with centrosome because of (a) the central role of the centrosome in replication process,[33] (b) the dependence of the processes of ageing and apoptosis[34] and gene expression[35] on cytoskeleton (in the formation of which the central role is played by centrosome),[20] and (c) information storage and reproduction ability of centrosomes.[4]

We have previously made assumptions about several possible mechanisms for determining most of the important cellular functions of centrosomal structures. We named one of them as an "RNA-dependent mechanism"; it combined two distinct mechanisms that we have provisionally named "cytoskeleton/centrosome-mRNA" and "centrosome-small RNA."[4] The idea regarding the participation of hypothetic centrosomal *small interference RNA* (siRNA) and/or miRNA in this process was proposed a year prior to the discovery by Alliegro and co-authors.[26]

It is acknowledged that siRNA and miRNA selectively inactivate expression of genes at a post-transcription level in cells.[36,37] Moreover, they participate in the regulation of expression of the most important genes in invertebrates and vertebrates.[38–40] Functions of small RNA vary from the formation of heterochromatin to destabilization of mRNA and translational control.[41,42] Small RNAs participate in almost every process, including identification of development time, cellular differentiation, proliferation, and death.[43]

It is also known that miRNA sequences are sufficiently conservative in related organisms,[44] whereas Alliegro et al.[26] made an assumption about the conservativeness of the cnRNA structure. On the other hand, conservativeness of all centriolar structures is also well recognized.[45]

If cnRNA is attributed to the class siRNA, miRNA, or *Piwi-interaction RNA* (piRNA)—the point being that dimensions of cnRNA exceed several times the dimensions of currently known siRNA, miRNA, and piRNA—the recent discovery of a large number of non-canonical small RNAs demolishes the

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notion of borders between small RNA classes,[43] and makes it possible to attribute cnRNA to the small RNA class in the future. Thus, it can also have some regulatory function. Its discovery may serve as another indirect verification of the correctness of our hypothesis. Moreover, the availability of this function in cnRNA is possible, considering that most of them contain fully sequenced cnRNA structures similar to reverse transcriptase.

We are not claiming that centrosome would not participate in the management of a cellular cycle and many other important cellular processes without cnRNA. Centrosome may affect cellular processes through cytoskeletal mechanisms, e.g. through a cilium,[46] by means of mRNA distribution[15] and other mechanisms (e.g., see our review[4]). Therefore, it is unlikely that our centrosomal hypothesis of ageing relates to "genetic theories" of ageing. Despite the fact that more than 200 genes relating to ageing have been discovered in various species of animals, many scientists consider that ageing is not caused by any special mechanisms; otherwise, spontaneous malfunctioning, e.g., through mutation, could easily lead to longevity and even immortality, which is not observed. Only diseases causing premature ageing are known.[47] Apparently the ageing mechanism (at least one of its forms—replicative ageing) is more or less universal for *Animalia*, which is a good argument against the majority of ageing theories.

Do research studies cited in our review verify the presence of an "intrinsic" RNA in centrosome rather than of a specific one? We believe that such a statement probably requires more data for verification. However, these studies have reliably demonstrated the existence of a closer connection between RNA and centrosome.

We emphasize that even the absence of its own "intrinsic" DNA/RNA in centrosome cannot be used as an argument against a hypothesis on their endosymbiotic origin. Today few researchers doubt the endosymbiotic origin of mitochondria that have lost 99% of their genome. Excess genes were lost entirely, whereas necessary ones were transmitted to the nucleus.[48] Mitochondrial DNA does not codify most of the proteins necessary for mitochondrion to function. Most mitochondrial functions are controlled by nuclear DNA.[49] If during the long process of evolution of eukaryotic cells mitochondria lost more than 99% of their genome, could the proto-centrosomes have lost it entirely?

ADDITIONAL REMARKS

Despite the fact that the aim of our article is to review studies on cnRNA, we considered it necessary to analyze other aspects of centrosomal functioning as well. Our article would be incomplete if we had not touched on some questions that can give evidence against the idea we had expressed about the importance of cnRNA participation in the regulation mechanisms of cell centrosomal functions. Facts on (i) de novo formation of centrioles in

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mammalian cells, and (ii) development and live insects in the absence of centrioles—do they contradict the hypothesis of centrosomal autonomy?

Briefly consider these aspects as well:

(i) De novo centriole assembly after centrosome removal by microsurgery or laser ablation has been observed in transformed[50,51] and normal mammalian somatic cells.[52] When centrioles are removed from even "normal" cells, new centrioles can form de novo with great efficiency.[46] Also, centrioles form de novo in parthenogenetically activated sea urchin,[53] Drosophila,[54] and to a limited extent in mammalian species.[55] In rabbits, de novo centriole formation is seen in parthenogenetic blastocysts.[56] Under the term "de novo" we imply the assembly of a new centriolar structure in the absence of a pre-existing one.[57] However, the absence of a pre-existing centriole does not in any way confirm that pericentriolar material (PCM) or centriole disassemble products are also absent in the cell. In most mammals, after fertilization, PCM eggs connecting the spermatozoid centriole ensure the formation of functional centrosome that is competent to generate embryo-specific microtubule formations for nuclear apposition, cell division, cell differentiation, and embryo development.[54,55,58,59] Oocytes of the animal species that lose centrioles contain materials for many rounds of centrosome assembly.[60] Data of Loncarek and co-authors[61] also suggest that formation of daughter centrioles is initiated within the PCM cloud. Therefore, researchers think that the reproductive ability of centrioles is not directly related to their triplet structure and that the centrioles may be formed beyond connection with pre-existing centriolar cylinders, i.e., the structure of centriolar cylinders per se is not necessary for the reproduction of this organelle.[59] Other authors also write about the "unrecognizable centriole substructures that provide the function of centrioles without the structure" (ref. [60], p. 14).

Considering this, could the de novo formation of centrioles in some cells be used to disprove the autonomy of centrioles and centrosomes?

There is no precise answer to this question, but it is interesting to mention that the formation of the de novo centrioles was one of those facts that generated the idea of the existence of nucleic acids in centrosome—according to many scientists nucleic acids are very well suited to the role of participants of replicative doubling of centrioles.[59] However, so far there is no evidence of the structural or regulatory role of RNA in the process of replication of these structures. And yet, the de novo centriole assembly can in no way be used to disprove the idea of centrosomal autonomy—at least because the centriole assembly proteins are localized to PCM.[18,62,63] Consequently, losing the

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"visible" part of the centrioles/centrosome does not at all mean losing of *all* of its components.

(ii) According to recent data, centrosomes are not essential for spindle assembly.[64,65] Acentrosomal cells can progress through mitosis, although their fate depends on the cell type and organism. In cultured monkey kidney cells, laser ablation of centrosome triggers cell-division arrest in the following G_1-S transition.[66] In HeLa cells, acentriolar cells possess a de novo centriolar assembly pathway that allows them to continue dividing.[51] These observations lead to the prediction that the absence of centrioles should have devastating effects during development.[67] However, Basto and colleagues[68] demonstrate that, at least in the Drosophila, this is not the case. In the above-mentioned study, they show that the Drosophila DSas-4 protein is required for centriole replication. DSas-4 mutant cells progressively lose centrioles during embryonic development and by third-instar larval stages, centrioles are essentially undetectable in these cells. These mutant flies develop at near-normal rates, demonstrating that flies can proceed through the majority of development without centrioles, centrosomes, cilia, or flagella. Mutant adults, however, die shortly after birth because they lack cilia in type I sensory neurons. Thus, centrioles are essential for Drosophila survival only because they are required for cilia formation.[68]

No doubt, these data neutralize to some extent the significance of centrosome for the cell. We note, however, that such data were received on Drosophila only; the centriolar structure in Drosophila and similar species basically differs from the centriolar structures of the majority of animal cells.[57,69,70] Could this be the cause of results received by Basto and colleagues?

In conclusion, we are once again (as in our first article) compelled to mention that the discovery of cnRNA confirms the correctness of Wheatley's opinion,[6] stated almost 30 years ago, that the centrosome/centriole complex remains a "central enigma of cell biology."

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